Advanced Hyperspectral Analysis

Endmembers, mixtures, and spectral matching

CSE 620B | Chapter 14.8-14.10

Quiz references: **Abundance Mapping and Analysis Techniques** (Week 10).

The Question Behind Unmixing

At 20 m spatial resolution, one pixel may cover several materials.

- Classification asks: which label best describes this pixel?
- Unmixing asks: what material fractions could explain its spectrum?
- Both depend on preprocessing, endmember choice, and model assumptions.

Endmembers and Convex Geometry

- Represent each pixel as a point in a space with one axis per band.
- Under a linear mixture, pixel spectra lie inside the convex hull of the endmembers.
- Endmembers form the vertices of a simplex: a line segment, triangle, or higher-dimensional analogue.
- Pure pixels are convenient vertices, not a requirement; algorithms can estimate vertices when no pure pixel is present.

Linear Mixing Model

For one pixel, the reflectance in each band is modeled as a weighted sum:

$$\mathbf{x} = \mathbf{M}\mathbf{f} + \boldsymbol{\epsilon}, \qquad f_j \geq 0, \qquad \sum_j f_j = 1$$

- $\mathbf{x}$: observed spectrum
- Columns of $\mathbf{M}$: selected endmember spectra
- $\mathbf{f}$: estimated fractions; $\boldsymbol{\epsilon}$: residual error

The constraints describe the fully constrained model; they are assumptions, not proof of a good fit.

A Small Mixture Example

Suppose a three-band pixel is a mixture of vegetation and soil:

$$\mathbf{m}_{veg} = [0.05, 0.10, 0.50]^T, \qquad \mathbf{m}_{soil} = [0.20, 0.25, 0.30]^T$$

With fractions $\mathbf{f} = [0.7, 0.3]^T$:

$$\mathbf{x} = 0.7\mathbf{m}_{veg} + 0.3\mathbf{m}_{soil} = [0.095, 0.145, 0.440]^T$$

The estimate is only as meaningful as the endmembers and mixing model.

When Linear Mixing Breaks Down

- Linear mixing models side-by-side materials as an additive weighted sum, without material-to-material scattering interactions.
- Multiple scattering, intimate mineral mixtures, and layered canopies can violate that assumption.
- A structured residual is evidence to inspect, not a reason to add complexity automatically.
- Nonlinear models can represent interactions, but require additional assumptions or calibration data.

Abundance Mapping quiz Q2, Q11: nonlinear mixing

Spectral Angle Mapping

Compare a pixel spectrum with a reference spectrum by the angle between their vectors:

$$\theta = \cos^{-1}\left(\frac{\mathbf{x}^T\mathbf{r}}{\|\mathbf{x}\|_2\,\|\mathbf{r}\|_2}\right)$$

- Smaller angles indicate more similar spectral shape.
- Positive rescaling does not change the angle, so brightness differences matter less.
- SAM returns similarity to a reference; it does not estimate mixture fractions.

Match, Classify, or Unmix?

Goal	Useful starting point	Main limitation
Match known material spectra	Spectral angle or library matching	Reference and scene conditions may differ
Assign one label per pixel	Supervised classification	Hides within-pixel mixtures
Estimate material proportions	Constrained unmixing	Depends on endmembers and mixing assumptions

Use the question and available reference data to choose the method.

A Practical Analysis Sequence

1. Confirm wavelength calibration, bad-band handling, and reflectance units.
2. Reduce noise and inspect retained MNF components.
3. Select or estimate candidate endmembers; record their provenance.
4. Fit fractions or compare reference spectra.
5. Inspect residuals and spatial patterns before interpreting the map.

Multiscale Features and Learned Models

- Wavelets describe local spectral detail at different wavelength scales: broad continuum shape versus narrow absorption structure.
- The transform alone does not identify a material or correct the atmosphere; discarding coefficients also discards information.
- A supervised model such as an SVM can learn nonlinear decision boundaries from labeled spectra.
- A nonlinear classifier is not automatically a physical unmixing model; validate its predictions on independent samples.

Notebook: Fractions Versus Spectral Shape

Run the spectral mixture demo in Colab

View the notebook as slides

Recover known fractions, then dim a reference spectrum. Which result changes: its brightness, its spectral angle, or its material fractions?

Check Your Interpretation

A pixel has a small spectral angle to a vegetation reference, but its estimated vegetation fraction is low.

- Are these results necessarily contradictory?
- Which assumptions differ between angle matching and unmixing?
- What evidence would you inspect before naming the material?